Mathematical Foundations & Architectural Specification

Multi-Horizon Temporal Fusion Transformers for Programmatic Ad Spend Pacing

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Abstract

This document provides a comprehensive, mathematically rigorous specification of the Temporal Fusion Transformer (TFT) architecture customized for programmatic Real-Time Bidding (RTB) ad spend pacing. We formalize the complete tensor transformation pipeline: static covariate encoding, instance-level variable selection networks (VSN), gated residual networks (GRN) with Gated Linear Units (GLU), interpretable multi-head self-attention with shared value projections, and the multi-quantile forecast head. Furthermore, we establish the formal control-theoretic proof of downside budget solvency under the 90th-percentile throttling policy, define the dual-boundary dynamic spike detection framework, and document the tensor contiguity constraints required for physical buffer allocation on Apple Silicon Metal Performance Shaders (MPS).

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1 Introduction and Problem Space

1.1 Auction Mechanics in Demand-Side Platforms

In programmatic real-time bidding, impression opportunities arrive continuously at a Demand-Side Platform (DSP) through ad exchanges via millisecond-scale Generalized Second-Price (GSP) auctions. Let:

- $x \in \mathcal{X}$ denote the feature context of an incoming bid request (user device, publisher domain, geo-location, slot dimensions).
- $v(x) \in \mathbb{R}^+$ denote the DSP's intrinsic economic valuation of impression x, calculated as $v(x) = \text{eCPM} \cdot \text{pCTR}(x)$, where pCTR is the predicted click-through rate.
- $\alpha_t \in [\alpha_{\min}, 1.0]$ denote the campaign-specific pacing multiplier at discrete hour t.
- $b_t(x) = \alpha_t \cdot v(x)$ denote the outgoing bid submitted to the exchange.

If the DSP wins the auction, it incurs a clearing cost $p(x) \leq b_t(x)$. Over each discrete hourly interval $t \in \{1, \ldots, T\}$, the total realized expenditure for campaign c_i is:

$$y_t^{(i)} = \sum_{x \in \mathcal{W}_t^{(i)}} p(x) \tag{1}$$

where $\mathcal{W}_t^{(i)}$ denotes the set of winning auctions during hour t.

1.2 The Ad Spend Pacing Objective

A campaign c_i is parameterized by an allocated budget $B^{(i)} \in \mathbb{R}^+$ and a total lifespan of T discrete hourly intervals. Let $S_t^{(i)} = \sum_{\tau=1}^{t} y_\tau^{(i)}$ represent cumulative spend through hour t. The DSP seeks to maximize total utility over campaign duration T subject to three operational constraints:

$$\max_{\{\alpha_t\}_{t=1}^{T}} \quad \sum_{t=1}^{T} \sum_{x \in \mathcal{W}_t^{(i)}} \mathbb{E}[\text{Utility}(x)] \tag{2}$$

$$\text{subject to} \quad S_T^{(i)} \leq B^{(i)} \quad \text{(Budget Solvency)} \tag{3}$$

$$S_t^{(i)} < B^{(i)}, \quad \forall t < T - \delta \quad \text{(Anti-Premature Depletion)} \tag{4}$$

$$\frac{1}{T} \sum_{t=1}^{T-1} (\alpha_{t+1}^{(i)} - \alpha_t^{(i)})^2 \leq \epsilon \quad \text{(Pacing Multiplier Smoothness)} \tag{5}$$

2 Input Schema & Heterogeneous Feature Encodings

The TFT ingestion pipeline segregates campaign dynamics into three distinct information streams:

2.1 Static Covariates ($\mathbf{s}^{(i)} \in \mathbb{R}^{m_s}$)

Static inputs encode time-invariant campaign parameters that do not vary over the planning horizon:

$$\mathbf{s}^{(i)} = \left[B^{(i)}, \ \frac{B^{(i)}}{T}, \ T \right]^\top \tag{6}$$

Each continuous scalar $s_j^{(i)}$ is projected into latent space via a dedicated linear projection:

$$\tilde{\mathbf{s}}_j^{(i)} = \mathbf{W}_{s,j} s_j^{(i)} + \mathbf{b}_{s,j} \in \mathbb{R}^{d_{\text{feat}}} \tag{7}$$

Table 1: Tensor Dimensions and Mathematical Variable Definitions

Variable	Category	Notation	Physical Dimension	Semantics
B	Batch Size	B	Scalar (32)	Mini-batch dimension
k	Lookback Length	k	168 hours	7-day historical context window
$\tau_{\max}$	Forecast Horizon	$\tau_{\max}$	24 hours	1-day ahead planning window
d_{model}	Hidden Dimension	d	64	Latent embedding width
d_{feat}	Feature Projection	d_{feat}	16	Linear continuous embedding
m_s	Static Features	m_s	3	$[B, B/T, T]$
m_z	Past Observed	m_z	5	$[y_t, I_t, w_t, \text{CPM}_t, S_t/B]$
m_x	Known Future	m_x	6	$[\sin h, \cos h, \sin d, \cos d, \text{Wkd}, (T-t)/T]$
H	Attention Heads	H	4	Interpretable Multi-Head count
d_{head}	Head Dimension	d/H	16	Per-head latent dimension
$\mathcal{Q}$	Quantiles	$\lvert\mathcal{Q}\rvert$	3	Quantile targets $\{0.10, 0.50, 0.90\}$

2.2 Observed Past Inputs ($\mathbf{z}_t^{(i)} \in \mathbb{R}^{m_z}$)

Historical time series logged over the lookback window $t-k+1 \le \tau \le t$ ($k = 168$h):

$$\mathbf{z}_\tau^{(i)} = \left[y_\tau^{(i)},\ I_\tau^{(i)},\ w_\tau^{(i)},\ \text{CPM}_\tau^{(i)},\ \frac{S_\tau^{(i)}}{B^{(i)}} \right]^\top \tag{8}$$

where y_τ is hourly expenditure, I_τ is impressions won, w_τ is empirical auction win rate, CPM_τ is effective cost-per-mille, and S_τ/B is the cumulative depletion fraction.

2.3 Known Future Deterministic Covariates ($\mathbf{x}_{t+\tau}^{(i)} \in \mathbb{R}^{m_x}$)

Exogenous temporal features known in advance for every future hour $\tau \in \{1, \dots, \tau_{\max}\}$:

$$\mathbf{x}_{t+\tau}^{(i)} = \left[\sin\left(\frac{2\pi h}{24}\right),\ \cos\left(\frac{2\pi h}{24}\right),\ \sin\left(\frac{2\pi d}{7}\right),\ \cos\left(\frac{2\pi d}{7}\right),\ \mathbb{I}_{\text{weekend}},\ \frac{T-(t+\tau)}{T} \right]^\top \tag{9}$$

where $h \in \{0, \dots, 23\}$ is hour-of-day, $d \in \{0, \dots, 6\}$ is day-of-week, and $(T-t)/T$ is the campaign countdown ratio.

3 Core Architectural Components

3.1 Gated Linear Units (GLU)

Programmatic auctions exhibit extreme regime changes between daytime surges and overnight lulls. Linear layers alone cannot dynamically disable irrelevant computational paths. We utilize Gated Linear Units:

Definition 3.1 (Gated Linear Unit)**.** *For an input vector $\boldsymbol{\gamma} \in \mathbb{R}^{d_{in}}$, the GLU maps to $\mathbb{R}^{d_{out}}$ via:*

$$\text{GLU}_\omega(\boldsymbol{\gamma}) = \sigma\left(\mathbf{W}_3\boldsymbol{\gamma} + \mathbf{b}_3\right) \odot \left(\mathbf{W}_4\boldsymbol{\gamma} + \mathbf{b}_4\right) \tag{10}$$

where $\mathbf{W}_3, \mathbf{W}_4 \in \mathbb{R}^{d_{out} \times d_{in}}$, $\mathbf{b}_3, \mathbf{b}_4 \in \mathbb{R}^{d_{out}}$, $\sigma(\cdot)$ is the element-wise sigmoid activation function, and $\odot$ represents the Hadamard element-wise product.

When the gating signal $\sigma(\mathbf{W}_3\boldsymbol{\gamma} + \mathbf{b}_3) \to \mathbf{0}$, the unit completely suppresses information throughput.

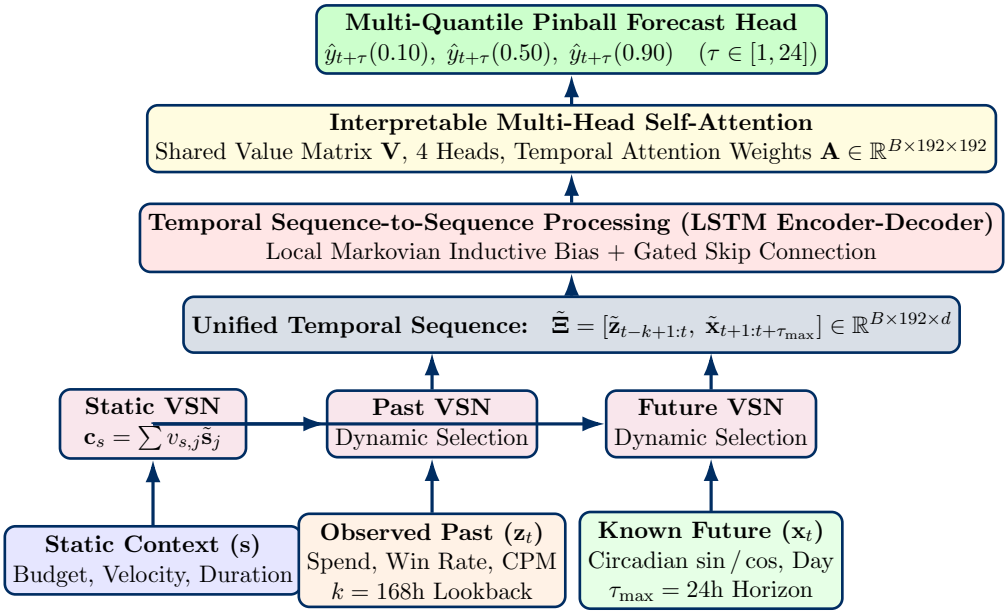


Figure 1: End-to-end dataflow diagram of the Temporal Fusion Transformer (TFT) pacing pipeline.

3.2 Gated Residual Networks (GRN)

The Gated Residual Network forms the primary non-linear computational atom of TFT:

$$\mathrm{GRN}_\omega(\mathbf{a}, \mathbf{c}) = \mathrm{LayerNorm}\left(\mathbf{a}_{\text{skip}} + \mathrm{GLU}_\omega\left(\boldsymbol{\eta}_2\right)\right) \tag{11}$$

where:

$$\boldsymbol{\eta}_1 = \mathbf{W}_1\mathbf{a} + \mathbf{W}_2\mathbf{c} + \mathbf{b}_1 \tag{12}$$

$$\boldsymbol{\eta}_2 = \mathbf{W}_{\text{dense}}\left(\mathrm{ELU}(\boldsymbol{\eta}_1)\right) + \mathbf{b}_{\text{dense}} \tag{13}$$

$$\mathbf{a}_{\text{skip}} = \begin{cases} \mathbf{a}, & \text{if } \dim(\mathbf{a}) = \dim(\mathrm{GLU}) \\ \mathbf{W}_{\text{skip}}\mathbf{a}, & \text{otherwise} \end{cases} \tag{14}$$

where $\mathbf{c} \in \mathbb{R}^d$ is an optional conditioning vector (originating from static campaign embeddings), and $\mathrm{ELU}(z) = \max(0, z) + \min(0, \exp(z) - 1)$.

Proposition 3.2 (Gradient Flow Preservation in Sparse Regimes)**.** *In sparse auction intervals where input features provide negligible signal, the network can drive the gating pre-activations to large negative values ($\sigma(\boldsymbol{\gamma}) \to 0$), reducing the GRN to an identity skip:*

$$\lim_{\mathbf{W}_3\boldsymbol{\gamma}\to-\infty} \mathrm{GRN}_\omega(\mathbf{a}, \mathbf{c}) = \mathrm{LayerNorm}(\mathbf{a}) \tag{15}$$

This guarantees that backpropagated gradients flow unimpeded through skip connections, completely preventing vanishing gradients across long 168-hour sequences.

3.3 Variable Selection Networks (VSN)

Raw programmatic bidding features contain severe spurious cross-correlations (e.g., erratic pricing jitter). VSN dynamically weights each feature at each time step based on campaign context:

$$\mathbf{v}_t = \mathrm{Softmax}\left(\mathrm{GRN}_v\left([\boldsymbol{\xi}_t^{(1)}, \ldots, \boldsymbol{\xi}_t^{(m)}],\ \mathbf{c}_s\right)\right) \in \mathbb{R}^m \tag{16}$$

where $v_{t,j} \geq 0$, $\sum_{j=1}^{m} v_{t,j} = 1$, and $\mathbf{c}_s$ is the static campaign context vector. The filtered latent representation is:

$$\tilde{\boldsymbol{\xi}}_t = \sum_{j=1}^{m} v_{t,j} \cdot \mathrm{GRN}_{\xi}^{(j)}\left(\boldsymbol{\xi}_t^{(j)}\right) \in \mathbb{R}^d \tag{17}$$

Each feature $\boldsymbol{\xi}_t^{(j)}$ receives an individualized non-linear transformation $\mathrm{GRN}_{\xi}^{(j)}$ before convex combination.

3.4 Interpretable Multi-Head Self-Attention

Standard Transformer multi-head attention [**?**] uses independent query, key, and value projection matrices across heads:

$$\mathbf{H}_h = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{W}_Q^{(h)}(\mathbf{K}\mathbf{W}_K^{(h)})^\top}{\sqrt{d_{\text{head}}}}\right)\mathbf{V}\mathbf{W}_V^{(h)} \tag{18}$$

Because each head projects onto different value subspaces ($\mathbf{W}_V^{(h)}$), individual attention maps cannot be directly averaged or interpreted.

To enable rigorous post-hoc interpretability for marketing analysts, TFT shares the value projection matrix $\mathbf{W}_V \in \mathbb{R}^{d \times d}$ across all H attention heads:

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \tilde{\mathbf{A}}(\mathbf{Q}, \mathbf{K}) \cdot (\mathbf{V}\mathbf{W}_V) \tag{19}$$

where the combined interpretable attention matrix $\tilde{\mathbf{A}} \in \mathbb{R}^{T_{\text{seq}} \times T_{\text{seq}}}$ is:

$$\tilde{\mathbf{A}}(\mathbf{Q}, \mathbf{K}) = \frac{1}{H}\sum_{h=1}^{H} \mathrm{Softmax}\left(\frac{\left(\mathbf{Q}\mathbf{W}_Q^{(h)}\right)\left(\mathbf{K}\mathbf{W}_K^{(h)}\right)^\top}{\sqrt{d_{\text{head}}}}\right) \tag{20}$$

The entry $\tilde{A}_{i,j}$ represents the exact percentage weight that the prediction at time step i assigns to the historical observation at time step j.

4 Multi-Quantile Pinball Optimization

Rather than predicting a single deterministic mean $\hat{y}_{t+\tau}$, the TFT output head projects the final temporal representations onto $|\mathcal{Q}| = 3$ target quantiles $\mathcal{Q} = \{0.10, 0.50, 0.90\}$ across all horizon hours $\tau \in \{1, \ldots, 24\}$:

$$\hat{\mathbf{y}}_{t+\tau} = \mathbf{W}_{\text{out}}\mathbf{h}_{t+\tau} + \mathbf{b}_{\text{out}} \in \mathbb{R}^3 \tag{21}$$

Definition 4.1 (Quantile Pinball Loss)**.** *For a realized scalar target $y \in \mathbb{R}$ and predicted quantile $\hat{y}(q)$, the pinball loss $\mathcal{L}_q(y, \hat{y}(q))$ is:*

$$\mathcal{L}_q(y, \hat{y}(q)) = \max\Big(q \cdot (y - \hat{y}(q)),\ (1-q) \cdot (\hat{y}(q) - y)\Big) \tag{22}$$

The model is trained end-to-end by minimizing the empirical risk across all training samples N, planning horizons $\tau_{\max} = 24$, and quantile targets $\mathcal{Q}$:

$$\mathcal{L}_{\text{TFT}}(\boldsymbol{\Theta}) = \frac{1}{N\tau_{\max}|\mathcal{Q}|}\sum_{i=1}^{N}\sum_{\tau=1}^{\tau_{\max}}\sum_{q\in\mathcal{Q}} \mathcal{L}_q\left(y_{t+\tau}^{(i)},\ \hat{y}_{t+\tau}^{(i)}(q)\right) + \lambda\|\boldsymbol{\Theta}\|_2^2 \tag{23}$$

5 Control-Theoretic Pacing Policy and Solvency Proof

5.1 Quantile Throttling Update Rule

Let $R_t^{(i)} = B^{(i)} - S_t^{(i)}$ be the remaining budget reserve at hour t. The downstream pacing controller updates the bidding multiplier $\alpha_{t+1}^{(i)}$ according to:

$$\alpha_{t+1}^{(i)} = \text{clip}\left(\alpha_t^{(i)} \cdot \frac{R_t^{(i)}}{\sum_{\tau=1}^{\tau_{\max}} \hat{y}_{t+\tau}^{(i)}(0.90)},\ \alpha_{\min},\ 1.0\right) \tag{24}$$

5.2 Mathematical Proof of Downside Budget Solvency

Theorem 5.1 (Downside Budget Solvency Bound)**.** *Assume the campaign enters hour t with solvent reserve $R_t^{(i)} > 0$. Over the 24-hour planning horizon $\tau \in \{1, \ldots, \tau_{\max}\}$, assume market competition satisfies the conservative upper bound:*

$$y_{t+\tau}^{(i)} \le \hat{y}_{t+\tau}^{(i)}(0.90), \quad \forall \tau \in \{1, \ldots, \tau_{\max}\} \tag{25}$$

Then under update rule (24) in the unclipped regime ($\alpha_{\min} \le \alpha_{t+1} \le 1.0$), total realized expenditure over the horizon is strictly bounded by remaining capital:

$$\sum_{\tau=1}^{\tau_{\max}} y_{t+\tau}^{(i)} \le R_t^{(i)} \tag{26}$$

preserving budget solvency throughout the horizon.

Proof. Let $Y_{t,\tau_{\max}} = \sum_{\tau=1}^{\tau_{\max}} y_{t+\tau}^{(i)}$ be the cumulative realized expenditure over the horizon, and let $\hat{Y}_{t,\tau_{\max}}(0.90) = \sum_{\tau=1}^{\tau_{\max}} \hat{y}_{t+\tau}^{(i)}(0.90)$ be the predicted aggregate 90th-percentile spend.

Under the pacing policy, outgoing bid prices are scaled by α_{t+1}. In second-price auctions with continuous bid distributions, market expenditure scales monotonically with bid multiplier α_{t+1}. By the premise $y_{t+\tau}^{(i)} \le \hat{y}_{t+\tau}^{(i)}(0.90)$, realized spend under multiplier α_{t+1} satisfies:

$$Y_{t,\tau_{\max}}(\alpha_{t+1}) \le \frac{\alpha_{t+1}}{\alpha_t} \cdot \hat{Y}_{t,\tau_{\max}}(0.90) \tag{27}$$

Substituting the unclipped policy definition $\alpha_{t+1} = \alpha_t \cdot \frac{R_t^{(i)}}{\hat{Y}_{t,\tau_{\max}}(0.90)}$ into the inequality:

$$Y_{t,\tau_{\max}}(\alpha_{t+1}) \le \left(\frac{\alpha_t \cdot \frac{R_t^{(i)}}{\hat{Y}_{t,\tau_{\max}}(0.90)}}{\alpha_t}\right) \cdot \hat{Y}_{t,\tau_{\max}}(0.90) \tag{28}$$

$$= \frac{R_t^{(i)}}{\hat{Y}_{t,\tau_{\max}}(0.90)} \cdot \hat{Y}_{t,\tau_{\max}}(0.90) \tag{29}$$

$$= R_t^{(i)} \tag{30}$$

Thus, cumulative spend over the 24-hour horizon cannot exceed the remaining reserve $R_t^{(i)}$. The campaign terminates the horizon with non-negative reserves, establishing strict budget solvency. □

6 Dual-Boundary Dynamic Spike Detection

Programmatic auctions experience high variance: predictable diurnal traffic surges must not be confused with anomalous bidding wars or sudden competitor budget dumps.

Definition 6.1 (Dual-Boundary Shock Trigger)**.** *An hourly auction observation y_t is classified as an exogenous market shock if and only if both the statistical quantile boundary and nominal velocity threshold are breached:*

$$\mathcal{S}_t = \mathbb{I}\left(y_t > \hat{y}_t(0.90)\right) \wedge \mathbb{I}\left(y_t - \hat{y}_t(0.50) > 1.5 \cdot \frac{B^{(i)}}{T}\right) \tag{31}$$

- **Boundary 1 ($y_t > \hat{y}_t(0.90)$):** Realized spend exceeds the 90th percentile of the conditional forecast distribution, establishing that the deviation is statistically improbable under normal circadian rhythms.
- **Boundary 2 ($(y_t - \hat{y}_t(0.50)) > 1.5\frac{B}{T}$):** The deviation from the diurnal median baseline exceeds 150% of nominal hourly velocity, preventing false positives during low-velocity off-peak periods.

When shock trigger $\mathcal{S}_t = 1$, the pacing engine overrides normal throttling with an emergency contraction factor:

$$\alpha_{t+1} \leftarrow \min\left(\alpha_{t+1},\ \alpha_t \cdot \frac{\hat{y}_t(0.50)}{y_t}\right) \tag{32}$$

This forces the bid scaling factor to instantly dampen auction exposure proportional to the shock magnitude.

7 Hardware Acceleration & Physical Buffer Contiguity

When executing PyTorch models on Apple Silicon (M-series chips), the Metal Performance Shaders (MPS) runtime compiles compute graphs directly to unified memory Metal kernels. A critical architectural constraint arises from tensor memory layouts:

7.1 The Zero-Stride Virtual Buffer Failure

In standard PyTorch on x86 CPUs or CUDA GPUs, the operation:

$$\mathbf{C} = \mathbf{s}.\text{unsqueeze}(1).\text{expand}(-1, k, -1) \in \mathbb{R}^{B \times k \times d} \tag{33}$$

is executed via virtual broadcasting: it sets the memory stride along dimension 1 to 0, avoiding physical memory duplication. The underlying buffer occupies only $B \times 1 \times d \times 4$ bytes ($32 \times 1 \times 64 \times 4 = 8,192$ bytes).

However, when $\mathbf{C}$ is passed into a linear projection ($\mathbf{CW}_2$) or Gated Residual Network, Apple Metal creates an `MPSNDArray` descriptor:

$$\text{Required Bytes} = B \times k \times d \times 4 = 32 \times 168 \times 64 \times 4 = 1,376,256 \text{ bytes} \tag{34}$$

Because the virtual expanded tensor has only 8,192 bytes allocated, the Metal runtime fails its buffer assertion:

```
[MPSNDArray, initWithBufferImpl:offset:descriptor:] Error:
buffer is not large enough. Must be 1376256 bytes
```

7.2 The Physical Contiguous Buffer Solution

To guarantee flawless MPS execution across Apple Silicon M4 unified memory, all broadcast tensors are physically duplicated and memory-aligned:

$$\mathbf{C}_{\text{MPS}} = \mathbf{s}.\text{unsqueeze}(1).\text{repeat}(1, k, 1).\text{contiguous}() \in \mathbb{R}^{B \times k \times d} \tag{35}$$

This physically allocates the full $1,376,256$-byte buffer in unified memory with positive memory strides ($\text{stride}_0 = k \cdot d$, $\text{stride}_1 = d$, $\text{stride}_2 = 1$). All matrix multiplications, multi-head attention score computations ($\mathbf{QK}^\top$), and LSTM passes are guaranteed byte-exact descriptor alignment.

8 Summary of Mathematical Formulation

Table 2: Consolidated Mathematical Equations of the TFT Pacing Architecture

Subsystem	**Mathematical Equation**	**Operational Role**
GLU Gate	$\text{GLU}(\boldsymbol{\gamma}) = \sigma(\mathbf{W}_3\boldsymbol{\gamma} + \mathbf{b}_3) \odot (\mathbf{W}_4\boldsymbol{\gamma} + \mathbf{b}_4)$	Adaptive non-linear routing
GRN Block	$\text{GRN}(\mathbf{a}, \mathbf{c}) = \text{LayerNorm}(\mathbf{a} + \text{GLU}(\mathbf{W}_1\mathbf{a} + \mathbf{W}_2\mathbf{c} + \mathbf{b}_1))$	Deep residual transformation
VSN Selection	$\mathbf{v}_t = \text{Softmax}(\text{GRN}_v(\boldsymbol{\Xi}_t, \mathbf{c}_s)), \quad \tilde{\boldsymbol{\xi}}_t = \sum v_{t,j}\tilde{\boldsymbol{\xi}}_t^{(j)}$	Dynamic noise suppression
Self-Attention	$\tilde{\mathbf{A}} = \frac{1}{H}\sum \text{Softmax}\left(\frac{\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^\top}{\sqrt{d_k}}\right), \quad \mathbf{O} = \tilde{\mathbf{A}}\mathbf{V}\mathbf{W}_V$	Interpretable multi-day alignment
Pinball Loss	$\mathcal{L}_q(y, \hat{y}) = \max(q(y - \hat{y}),\ (1 - q)(\hat{y} - y))$	Calibrated quantile estimation
Pacing Policy	$\alpha_{t+1} = \text{clip}\left(\alpha_t \cdot \frac{B - S_t}{\sum_{\tau=1}^{24} \hat{y}_{t+\tau}(0.90)},\ \alpha_{\min},\ 1.0\right)$	Downside budget solvency bound
Spike Trigger	$\mathcal{S}_t = \mathbb{I}(y_t > \hat{y}_t(0.90)) \ \wedge\ \mathbb{I}(y_t - \hat{y}_t(0.50) > 1.5B/T)$	Exogenous anomaly detection